PREPVECTOR
Cognitive Assessment Platform

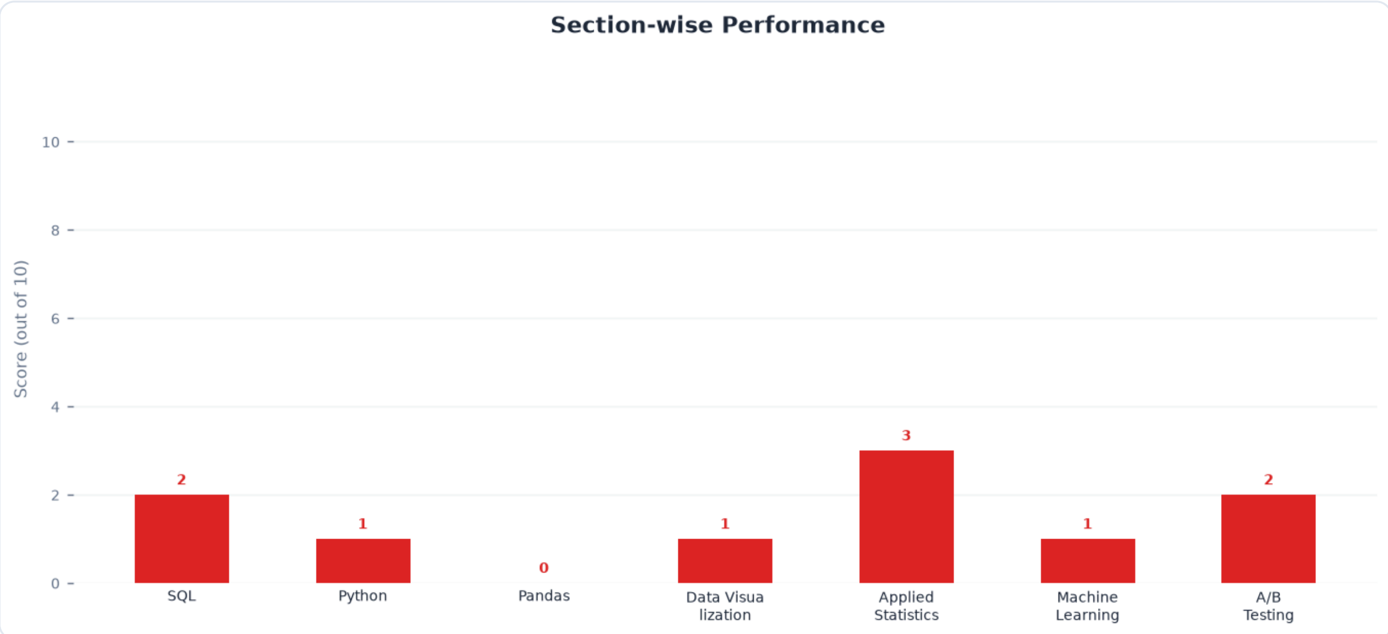
AI Evaluation Report

Data Science Expert — Self-Assessment

CANDIDATE	FINAL SCORE	CORRECT ANSWERS
Ritam Sur	**21** / 70 weighted points	**10** / 35 questions NEEDS IMPROVEMENT

Assessment Date: July 25, 2026 **Report Generated:** July 25, 2026 at 19:01 UTC **Role:** Data Science Expert

OVERALL RATING	SUCCESS RATE	WEIGHTED SCORE	TOP SKILL	NEEDS IMMEDIATE ATTENTION
★★☆☆☆	**30.0%**	**21 / 70**	**Applied Statistics**	**Pandas**



Skill Rating & Benchmark

Section	Your Score	Reference Benchmark (Demo)	Delta	Status
SQL	2 / 10	6.8	-4.8	Critical Gap
Python	1 / 10	6.2	-5.2	Critical Gap
Pandas	0 / 10	5.8	-5.8	Critical Gap
Data Visualization	1 / 10	6.0	-5.0	Critical Gap
Applied Statistics	3 / 10	5.5	-2.5	Needs Work
Machine Learning	1 / 10	5.2	-4.2	Critical Gap
A/B Testing	2 / 10	6.4	-4.4	Critical Gap

Attempt Statistics Breakdown

Section	Total Q	Attempted	Correct	Incorrect	Skipped	Accuracy
SQL	5	5	2	3	0	40%
Python	5	5	1	4	0	20%
Pandas	5	5	0	5	0	0%
Data Visualization	5	5	1	4	0	20%
Applied Statistics	5	5	3	2	0	60%
Machine Learning	5	5	1	4	0	20%
A/B Testing	5	5	2	3	0	40%

Executive Summary

The candidate achieved a correct-answer count of 10 out of 35 (28.6% accuracy) with a weighted score of 21 out of 70. This performance indicates significant foundational gaps across core data science competencies, particularly in SQL, Python, Pandas, and statistical analysis. Immediate, structured upskilling is required to meet industry benchmarks for technical data science roles.

Key Strengths

- **Conceptual Exposure:** Familiarity with advanced concepts such as FIFO inventory calculations, network interference in A/B testing, and PCA scree plots.
- **Tooling Awareness:** Recognition of advanced Python modules (like `itertools`) and complex Pandas operations (such as `merge_asof`), showing a solid theoretical awareness of the modern data science stack.

Top Blind Spots

- **Core Programming Syntax & Logic:** Gaps in fundamental execution, including basic SQL query structure (`ORDER BY` syntax), Python collection properties, and basic Pandas boolean indexing.
- **Statistical Inference & Experimental Design:** Difficulty interpreting basic hypothesis tests (p-values), handling ANOVA assumption violations, and identifying standard A/B testing errors and sample size drivers.
- **Data Visualization Best Practices:** Lack of familiarity with Matplotlib's object-oriented API, handling high-cardinality overplotting, and optimizing vector export performance.

Actionable Learning Resources

Complete the "SQL Zoo" interactive tutorials, specifically focusing on the `GROUP BY`, `SUM`, and Window Functions sections.

✓ Action Item

Read Chapters 5 and 8 of *Python for Data Analysis* (3rd Edition) by Wes McKinney to master Pandas indexing, MultiIndex hierarchies, and asynchronous merges.

✓ Action Item

Study Chapter 3 ("Linear Regression") and Chapter 4 ("Classification") of *An Introduction to Statistical Learning (ISLR)* to reinforce hypothesis testing, ANOVA, and model evaluation metrics.

✓ Action Item

Practice Matplotlib's object-oriented interface and density visualization techniques using the official Matplotlib "Lifecycle of a Plot" tutorial.

✓ Action Item